Active Inference BookStream #001.09 ~ "Governing Continuous Transformation"

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foreign 2023 we're here in active bookstream number 1.09 off to you blue hi I'm blue Knight I am a researcher in New Mexico and we are the active and Friends Institute of participatory online Institute that is communicating learning and practicing applied active inference you can reach us at our social media links here and um just a reminder this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and we will be following good video etiquette for live streams and just as we're mentioning that all backgrounds and perspectives are welcome here we will be having our final discussion coming up in about four weeks so if you would really like to participate if you've watched some of the live streams and have some questions we would really like to get some outside perspective so please get in touch with us so that you can participate in our final discussion um so like I said if you want to participate this is how uh and so we are still discussing governing continuous transformation um and today our goal is to discuss section nine so chapter nine um and we will be doing book map road map and all of the typical things so I already introduced myself um Daniel just gave us the date why don't you introduce yourself it's Daniel in California I'll pass to Tyler uh I'm Tyler in California I am a uh Dao researcher and protocol designer all right blue all right so here we are at the end of part two so next week uh the author is going to join us for a discussion over part two the free energy governance section of the book and then we'll be moving into part three I think we're gonna do two chapters at a time as we go through part three and I don't think we'll be discussing the engaged scholarship interviews but if you're curious about them I encourage you to read the text okay uh Tyler do you want to read this first part of the abstract for us sure all right so beneath all chaos is an underlying structure a challenge is to exploit disorders underlying order survive living organisms are encoded to resist disorder I.E entropy by way of self-organization firms are nothing more than molecules and are subject to obeying the very same laws thermodynamics entropy will eventually rise three specific propositions previously developed as part of this book's underlying dissertational research are outlined one structure the stronger affirms cross hierarchical top down bottom-up cognitive throughput coupling between and within governance and operational channels the less dominant firm's top-down approach cultivating a form of CEO centrism and strategy ownership all right blue you want to take the next next one yes so two cognition the stronger affirms cognitive disposition toward environmental enactment and Innovative abduction as opposed to adaptation the stronger the firm's strategic control and the smaller the risk of falling for the financial control trap three capabilities the stronger the board's dmn Centric Duality stewardship empowering upper echelon to sustainably balance exploitation versus exploration as well as Financial versus strategic control the stronger the sustainability of successful and continuous strategic renewal and hence the firm's prospects for survival strategy is less macroscopic but more subatomic less linear and deterministic but more Quantum and discontinuous there is a dominant tendency of glorifying the realest beauty of top-down linear cause effect relationships to the very detriment of quantum self-organization and I'm going to start off just reading this um heart from the book so Bijan says I'm trying to see it over my screen order anywhere in the cosmos is so singular it always requires an exemplary mechanism or process while randomization requires no I can't read the word it is simply The Way of the World when particles are allowed to act randomly whether it's the contents of a glass on of club soda on the rocks or the countless atoms making up the air in the room they slam into each other exchanging energy into complete randomizations of their positions and velocities is observed and so Bajan quotes um this lonza at Al 2020 paper and I was just like thinking of like firms as like organized organizations of people and like it really brought this picture to mind of like you know Adam's randomly slamming into each other like people randomly like slamming into each other like the mosh pit versus like the board meeting I just wanted to bring this like a juxtaposition of like like what is it to like minimize free energy in terms of like assembling humans as opposed to like assembling atoms anyway I just had to share with this like mental image um okay Tyler do you want to read this one for us sure all right so beneath all chaos is an underlying structure our challenge is to exploit disorders underlying order to survive living organisms are encoded to resist disorder or 90 entropy by way of self-organization firms are nothing more than molecules and are subject to being the very same laws thermodynamics uh and so yeah I thought this was I thought the connection here that was interesting was comparing uh saying that the laws of thermodynamics apply to firms and that's what's really going on and we've talked about this in previous streams I know we have uh Blues kind of talk about a little more in detail here about whether that's more of like an analogy comparing to thermodynamics or something like a real underlying structure of thermodynamics and we'll talk about that more shortly that also brings to mind like the mosh pit like and like I wonder in our base case as humans like are we like just chaotic like is our natural inclination to be like increasingly chaotic and like we override that um it just I don't know I I had all these mental images as I was reading this as well so um oh yeah Tyler uh put this in so I love this um I'm gonna read it because I love it uh so Vision says oh he actually quotes Gould at Owl 2016. so um and in the quote he says if physical theories were people thermodynamics would be the village witch over the course of three centuries she smiled quietly as other theories Rose and withered surviving major Revolutions in physics like the Advent of General relative relativity and quantum mechanics the other theories find her somewhat odd somehow different in nature from the rest yet everyone comes to her for advice and no one dares to contradict her Einstein for instance called her the only physical theory of universal content which I am convinced that within the flame framework of applicability of its basic concepts will never be overthrown her power and resilience lay mostly on her Frank intentions thermodynamics has never claimed to be a means to understand the mysteries of the natural worlds but rather pass toward a efficient exploitation of said World um yeah do you want to add comments here yeah I mean I liked about this was that like the kind of like mystical business leader like a Steve Jobs character is kind of like a Trope in the business world and it's kind of like almost like a joke at this point but there's also like a real case for the mystical business leader which is almost kind of what I think the John is like describing here is that you know the real state of the world of like operating um as a firm in a very uncertain uncertain is probably not the word I should use but like inactive in an active sense like is a very mystical receptive way of engaging with the world um and I really like making that connection of mysticism with like business with like you know corporations one other angle there is organizations are doing work they're doing hopefully useful work and thermodynamics is concerned with how thermal energy that uncorrelated exchangeable entropic dissipation can be channeled into moving the steamboat forward or lifting up the weight on the crane and so sometimes with the amount of seething activity in formal firms or informal groups it's like but if we all just went in this exact Direction at this exact time we could do this it's like well if all the air molecules just slammed against the wall it would fall over or if they all formed into uh a droplet you'd get rain in the room like thermodynamics helps us understand what is likely to happen given those interacting State spaces and again interestingly it has to do with extracting work useful work from this background of just vibration that's not work yeah yeah and so the other thing I like too just to build on that is that like it's okay to not know what's going on is that like that doesn't necessarily need to be a source of stress it's just like you can harness those energies and kind of see which way the wind is blowing so to speak without necessarily understanding understanding all the underlying principles I'm like that's totally okay and I think that's generally something that gives a lot of people both in the business world and even in our personal lives a lot of stress but this is also I think a very like freeing way to approach uh an uncertain and uncertain world yeah and and there's also like um just come maybe to close the loop or maybe open it up a little bit more um there's also the prospect of doing work that you don't know is going to be fruitful right and so like that is this like uncertainty minimization like you could spend you know your whole like an hour trying to crack open a nut and the nut is empty like like that's also possible right so like doing even though you're doing work like that's maybe useful work um it might not bear fruit so there's there's that like uncertainty element like as just a human or also like as a firm so I just wanted to bring back these slides like while we're talking about thermodynamics we talked about this a long time ago and we talked about the difference between like thermodynamic entropy and information theoretic entropy um and so I just wanted to kind of bring this back and just remind everyone that um there's a difference uh there is relationships between the two that are still being hashed out like mathematically um and and not totally understood but but the second law of Thermodynamics is that this entropy like disorder in the universe is always increasing um and like here like the you know the energy is always lost in a transfer in in work as heat to the environment in yeah so less and less usable energy becomes available all right um and then in information Theory we kind of talked about entropy as like with an inverse relationship with information so here you can see like a event that is low probability is high information so like if you're decoding a stream of letters I always like to think about it like this because it was explained to me like that and it made so much sense if you're decoding a stream of letters and and like you're expecting a message and this was Claude Shannon um and kind of like how he or how it was explained to me how how he kind of came up with this so you're decoding a message and you get the letters t h it would be really freaking weird if like the next letter was Z like I don't know any words that have THC like that's a very low probability event right so so it would be like a very high information thing or like you know maybe something came before the th and you missed it but there's some probability like t-h-e would be great or like t-h-i but like th Z wouldn't make any sense unless like the th was at the end of some previous word and so like you can kind of deduce what to expect in those kinds of situations um and so yeah the um probability distribution is like when there's 50 50 probability so so this is the second graph on the right when there's 50 50 probability like the flip of a fair coin that's super high entropy like you don't have any idea if it's going to be heads or if it's going to be Tails um but like it here it's low entropy if it's like a coin with two heads there's heads on both sides like you can only come out with heads so like there's very low entropy and such a coin flip because the outcome is always heads um right if there's no distinction between the two sides so I don't know that's the thermodynamic entropy versus information theoretic entropy and I think we talk a lot about thermodynamic entropy but maybe forget that the fep is really kind of founded in the information theoretic entropy as well um do you guys have anything to add here no okay um all right so Bhajan makes this claim in this section and I'm not going to read the claim but I'm just going to point out that here we kind of have a really concise definition of like he's been emphasizing the structure cognition and capabilities for a few chapters now and so here he defines structure as the the circular coupling circular coupling of governance predictions and operational stimuli channels in the form of prediction error minimization he defines cognition as environmental engagement in the form of enactment rather than adaptation and he defines capabilities as sensing and sense making in the form of Duality management and so this chapter was really kind of a good um like summary chapter I thought I can and kind of gives us a chance to go back and revisit like a couple of the concepts that that we talked about before but maybe need to highlight one more time do you want to read this one Tyler Frost sure so yeah through the rest of the the chapter John is going through each treasure of cognition um and he's really kind of going deep and kind of recapping what he's been covering through the previous the rest of the book uh so for structure he says the stronger affirms cross hierarchical top down bottom-up cognitive throughput and operational channels the stronger the firm's orientation towards free energy surprise minimization based on self-organization the less dominant affirm's top-down approach cultivating a form of CEO centrism and strategy ownership the stronger sustainability of successful and continuous strategic cool thanks um and so this is kind of like the the punchline of the structure and so I pulled out maybe just a couple topics from um from before the punch line uh to talk about um and so this kind of like top-down bottom-up coupling I was like looking for a picture of like what just top down bottom-up coupling like look like like how do you see that can I visualize that actually I was like looking for a picture to put on the previous slide and I came across this like 2008 paper from emotion residence to empathic understanding a social developmental Neuroscience account and like my background is neuroscience but it's definitely like cellular and molecular Neuroscience so I don't really have a good like comprehensive reading of of Social and developmental like neuroscience and what that looks like and I was like I loved this figure because it just was so like encompassing of so many of the things that um we've talked about so far Daniel have you read this paper have you ever seen this before like I hadn't either and I came across the figure and I was like oh I have to get the paper now um and so it was nice to kind of just delve into this and it's really about um what's required for empathy um but I just is so and I'll talk more about it but here you see like the top down bottom up coupling um so it's bottom-up information processing top-down information processing relation to like with really relative to the individual and then the context or situation um and and there's this perception action coupling and then like 2008 I'm like the perception action loop from 2008 this is cool um and and they talk about shared representation um and this like self other awareness which we talked about a couple book streams ago like um how like distinguishing the the self versus The Other Self and non-self um so I thought that this was really and executive functions regulation and control metacognition like the looking at um like that's like double Loop learning the the metacognition aspect so I thought that this paper was really cool and and um it really was like wow all of this is is like grounded in empathy um anyway it was me it was a good read uh it's Destiny and Meyer of 2008 and the link is up there if you want the DOI or any of that stuff um and it made me think like all this is grounded in empathy so like what is like we know like or at least it's thought in business that like to be successful as a business person your emotional potion your EQ is like way more important than your IQ like your ability to relate to others and interact with people so like a good emotional High emotional quotient versus like an intelligence quotient is way more crucial to um business success and so I just wonder about like the relationship between if any between free energy governance and like the emotional quotient like is there like I like John never talked about this directly in the book but it just made me think like how important is empathy for this kind of governance structure um I don't know if you guys have any thoughts here on this yeah I mean I think empathy is just really like sensitivity to your environment or sensitivity to like other actors and their internal States and so in the context of a firm like it doesn't necessarily need to be like their emote the you could think of an emotion necessarily as like for a another firm uh maybe the internal state of this external other firm so maybe like you are dependent on this other firm things aren't kind of going the way you would expect them to go with this other firm maybe orders or a backlog or their API is being kind of intermittently down maybe that is like a signal into the internal state of that firm and the internal state of the ecosystem as well and so I think that that might be like analogous to an emotion in a human context Maybe they had a bunch of definitions actually in the paper and I can't pull it up because I'm I'm screencasting so I'm not going to let you guys like go through my go through all my 25 000 open windows but maybe we'll pull them out um and and this is like maybe a good question for Bhajan about the relationship between empathy and the free energy governance when we get to see him next week but they had like several different definitions of empathy too in the paper which I thought was interesting they really really tried to take like a very like logical approach to it and it's cool um okay so here is a claim from the paper Bhajan says the higher the organizational friction in terms of obstructing bottom-up error messages traveling to and reaching top-down prediction generators the higher free energy the potential for surprise and the more likely that top-down dominant Logics fossilized prediction models and hence reinforce a firm's entropy um and like this is something that maybe Tyler and I have brought up either on stream but we've talked about it off stream a lot like how do we reduce this organizational friction like where are the um examples to that show us like what is the best way to like reduce this friction to like create a good top-down bottom-up coupling like we know that this coupling is super important like it all layers across the hierarchy but like can we find like an empirical way like that works in all situations to build that that structure um and that that I think is like a challenge or maybe a future research question I guess another thing that is a little counterintuitive about this too is that oftentimes the things that will reduce friction and maybe like the longer term feel high friction in that moment so for example like in previous streams we've been talking about like dissonance as being like a way of thinking about that friction right well the easiest way to deal with dissonance the lowest frictional way you could deal with it personally is just like not thinking about it and just like tunnel vision and just like ignore it and don't worry about these weird feelings that you have um and that but that obviously doesn't help the version of the organization as a whole um and so yeah there's this kind of especially in your personal life too by analogy is like confronting your own emotions that you don't want to think about in the short term really easy to just like power through and just keep doing what you're doing but if you can like process these and integrate these emotions um that's a lot healthier for you long term so it's an interesting kind of like tension between like your individual Tendencies of like not winning friction and like friction in the organization as also to connect back to the thermodynamics what is friction well it's when a force is applied on some surface or through some media and it results in heat which is dissipated taking away energy from what could have been done for work and now think about friction engineering do we coat every object with Teflon do we want everything to just at the lightest touch fly off into space so just because we can increase or reduce friction actually doesn't tell us especially over time scales like you're bringing up Tyler whether we want something to have a lot of friction like where the rubber hits the road or whether there are some fundamental trade-offs with how much heat the tire is losing in contact with a smooth Road versus how much grip it has on a loose Road and so all these different trade-offs are where the actual Art and Science is going to come into play but friction is not just a term that's used every day to describe annoying communication patterns FEG is pointing towards that potentially being understood as an actual physical Force cool uh do you want to read this one for us Tyler sure so the second proposition in cognition so the stronger affirms cognitive disposition towards stronger the firm's orientation towards free energy surprise minimization based self-organization the stronger the firm strategic control and the smaller the risk of falling for the financial control trap the stronger the sustainability of successful and continuous strategic renewal and hence the firm's prospects for survival all right and so I I I'm gonna read another quote from this section on cognition um and I brought back this picture because I think it's important um okay so Bijan says cognition in the form of world enactment is the purpose-directed action-centric and social construction of shared meaning and shared expectations anchored in the understanding that a firm's environment is a matter of environmental engagement in the form of interaction revealing significance and assigning probabilities to a bounded set of external unknown States and so like I'm coming back to this paper and I'm looking at like the information processing bottom up top down and like here I see this like shared representations and we've seen this like in like in this figure in this paper they're talking about like body language and like somatic response like heart rate increase or lower and all these things that have been measured in like the generation of empathy but like is is empathy maybe something that is required for shared meaning or shared expectation like I I wonder is like do you have to have that like like it is a sociopath like devoid of empathy empathy like living in a world where like things don't mean the same thing and expectations aren't the same like maybe right like there might be a a strong link there that I wonder if it's been explored or not so I'm just curious if you guys have thoughts on what developing shared meaning and expectation is and if it requires some kind of empathy with others yeah I I share that feeling one of the big questions I've had through this and honestly it's on a personal level

I've always been interested on like whether you can build tooling that would help create that shared meaning and expectations uh or whether this is really just constrained by like the individual like if you're if you have people who are sociopaths in your organization um maybe it's just like there's certain the way that they're wired it kind of limits the way that they can engage and create shared meaning with the rest of the organization and so yeah I'm kind of curious what you all think about that too like I don't have an answer for it but then maybe that'd be a good question for Sean have interpersonal empathy as a behavior as a feeling that's one whole area but in the creation of organizations it's almost like we have to ask not how strongly we feel or what we feel but whether some future person in that organization outside of our Direct perception cognition action Loop is going to have some sort of outcome because I think it's easy just to sketch out that you could feel really empathy empathetic or any emotional state and it could result in a totally discordant State even for somebody you care about or vice versa so how do we not get lost with just the introspection of the feeling of empathy and translate that into systems whose efficacy is equivalent to what we would have intended but that may not feel like empathy all the time yeah I mean I feel like especially for the last couple chapters we keep running to like a similar concept of like Earth theme of how the individuals wired in terms of like you know empathy and their ability to create shared meaning and how that bubbles up into the rest of the organization um and yeah like I keep feeling like there's no way out of like hey we're just limited as humans in our ability to integrate external information and that limits the ability of the organization but sounds like Daniel was maybe sounding more optimistic now like well actually maybe you can build some maybe there's tooling and systems you can build that can have organizational empathy even if the individual doesn't have it so that brings to like like something that Daniel and I have talked about before um working on like the discussion of memes like randomly random insertion random plug for the Eco meme project um but like when you're really thinking about painting a picture to convey like a shared meaning like we talked about the Echolocation how the dolphin like literally can like paint a picture like I saw a fish over here and can like translate like puts the picture of the fish into the other dolphin's mind like there's the picture of the fish right and so like you literally can paint the entire picture so like even if we could do something as remarkable as that um we convey only the semantic content like if if you can say like the fire hydrant is red right and like you guys both know like can can picture it in your mind like a red fire hydrant right but like we don't know what semiotic content is contained in that message like maybe someone's colorblind maybe they don't know if it's red or green like they can't tell right and so like the fire hydrants appears in this gray thing that they think might be red or maybe like someone has been in a house that burned down and the fire like the Hydra didn't work and they have this like trauma inspiration moment right like so so the semiotic content the symbolism behind the message or the picture that you painted even in a situation where you can perfectly create like the tooling to come to the shared meaning or representation like you're never gonna get the symbolism that exists for someone else it's a highly personal thing like we can have shared symbolism too culturally for sure but there's a deep level of personal symbolism in every in every human that's dependent on your previous life experience I think ah so coming back to abduction because um we did talk about it in the section I can't remember if it was in the last quote or not but I just wanted to um point out this is a recycled slide but coming back to abduction where Bajan talks about it is you know abductive reasoning this is a quote from the book abductive reasoning usually begins with a surprising observation or experience this is what shatters our habit and motivates us to create a hypothesis that might resolve the anomaly abduction is an inferential procedure in which we create a conjecture that if it were if it would make the surprising anomaly part of our normal world of our normal understanding of the world so like this picture is like the best way for me to always think about abduction like you can have a specific observation and you can come to a general conclusion right and then you can have deductive reasoning where you have a general rule but that's like leads to a specific conclusion that's always true and then when you have like an incomplete observation like you can't see what's going on you just go to the best guess and so in business and Bhajan talks about this again and again and again like you have to use abductive reasoning because we're never going to have a complete picture of like the past and future of the market like the whole market right or even your target audience so you're dealing with um the best logical guess um oh okay do you want to read this one for us Tyler this is you sure uh so the firm's generative model is the firm's Eco Niche if uncertainty and unpredictability dominate our worldly perception well then this reflects the very limitations and corrosiveness of our models and the underlying processes generating those models there's no objective uncertainty there's only your uncertainty environment is inactive each Act opens a new Fork of multiple paths leading to to potentially very different outcomes realities subjective algorithm applying an infinite number of worlds in superposition our actions not only create the junctures leading to different paths but we are in control to decide which path to take at each uh so yeah I thought this really summarized well um the kind of inactive cognition so there's a proposition still in proposition two here um of FEG and also that was like a fun quote because we talked about a lot on the last stream but uh John's been using kind of quantum uh language here and we talked about that on the last stream of K what exactly does does he mean there by that blue Daniel anything to add here I love your Excel spreadsheet Quantum superposition of an Excel spreadsheet because that's great it's capturing the late night blurry what action selection to take based upon the spreadsheet great all right so Bhajan makes this claim also in this section he says in fact we are challenged not to allow for the human brain's conceptual limits to solely determine perception and sense making instead Outsourcing some of the heavy information lifting to the environment in the form of extended cognition is the essence of optimizing prediction error minimization the more exponential and hence discontinuous a market environment the more we need self-organizing mechanisms that leverage the world as a supervisory signal challenging in circular causality the very perceptual content of our predictions and so like we've talked a lot about like cognitive offloading or extended cognition in terms of like cumulative cultural Evolution not in this bookstream per se but like over the course of active inference like from the very first beginning live streams we've talked about it a lot um because you know we now our so much thinking is done um externally like in it's encultured at at some point like but with the smartphone like we don't have to know how to get anywhere anymore we'll just map it so I I just I'm thinking like that this self-organizing mechanisms that leverage the world as a supervisory signal challenging in circular causality it's a very perceptual content of our predictions like I mean I'm so amazed by Chachi petite like I'm using it so much um just because it's like what does everybody know like chat GPT tells you what everybody knows it is like the selective synthesis of all information right and so like if you know something that chat gpt3 doesn't know like that's great you can write a blog about it but now like chat gp3 you can just write all the blogs about every other thing because there's just like no like it knows everything right it's got a it's got It's got all of the information synthesis except like the rare pockets of like knowledge so if this is this this self-organizing mechanism this chat gpt3 like a global search synthesis information management system I don't know what do you guys think about that well this is very out of my depth both in terms of like machine learning and active inference uh the best thing about this last night because this is like a huge debate right now it's like whether tragedy T3 is actually showing like emergent intelligence or whether it's just like a clever clever algorithm um and I was thinking about this too like what are the differences between like chat gbt3 and what I've been learning so far about active inference and like The crucial things it seems to be missing is like the inactive aspect of it like it's not really in dialogue with this world it's like stringing together information that's like aesthetic right it doesn't even incorporate new information like if you were to ask it about the news yesterday it couldn't it couldn't do that and it's not making it's not taking actions in the world and getting feedback it's just like the static model of how the world works and so I don't know it feels like fundamentally different to me and I'm not sure what that means I'm not sure if like something has to display active inference to have true intelligence for this is like a big hotly contested topic I'm sure you guys have many live streams about this uh but yeah Daniel blue I'm curious what you guys think well in the space of communication the response it provides to you is an action that's not moving a mechanical arm on a table but in our internet of everything it's not so hard to see how the digital outputs from a generative algorithm do translate to physical objects moving but it may not even need to do that it can send out messages on social media and definitely meriting further discussion as things will continue to develop and gain more access to recent information World model learning and action capacity soon so like I I have a a little bit of a of a like technical response here I think that the question is not about true intelligence maybe but but the question that I've I hear talked about a lot is sentience right and so but like I had read this big article about like is that is gpt3 sentient and I think it was like from one of the inside Google people the one that was released and now they're talking about it because everyone's using it and like sentient they just said like you know able to think and act independently but that's like not the definition of sentient like I'm not sure what the definition of sentient is but it's not like autonomous like they're using sentient and autonomous the same way and sentient is like able to feel things right like so I think like let's look it up right now let's Define sentient for For The World At Large what does that mean because intelligent sentient and autonomous are all different so hold on I'm looking on my phone because I'm screen sharing words words words Alice in Wonderland people will use them how they want but that's everything so as sentient is able to perceive or feel things right and so it's gpt3 healing anything I highly doubt it um perceiving a lot sure um but as to your thing about updating Tyler like I'm sure the model will update at some regular interval so like just think about um you know I mean we're not there yet in terms of compute power but maybe in 10 years like it will be training on the entire Wikipedia catalog every single day so like does Wikipedia have like a date entry for every day I think it does right so like when you ask like what was the news yesterday it'll know that like that's just a matter of what's encoded and what what's being processed through silicone it's only time right like we're we're way close so like it will be able to update um at some with some regularity like will it update itself I don't know that that gets into like autonomous acting I don't know those all those things are different I mean that reminds me those because we've had this discussion over the past you know 15 minutes about like empathy and the new organizations have empathy right and it's kind of like the same conversation it's like gpt3 of like okay well like maybe organizational empathy like it looks and just feels differently than like individual empathy right um and so you can imagine the same the same thing happening in gpt3 and I don't know what that version of empathy would look like in gpt3 but maybe it's just like something that we can't quite wrap our hands around it'll be awesome again I think a further discussion would be very Timely especially if for example communication from one person's syntax could move into a semantic space or a rhetorical space from the Eco meme direction as blue mentioned and then be translated back into another semantic space for the listener that might be changing the amount of math detail or in a different human natural language so there's a lot of incredible opportunities yeah definitely I think um you know it'll be interesting to see to interrogate Bhajan with such like esoteric questions to see to interrogate Bhajan with such like esoteric questions and see what he thinks like does a firm have empathy like what where did you even get this well we got it sorry we're off to ask um anyway all right so moving on uh Tyler do you want to read this one preposition sure so the stronger the board's dmn Centric Duality stewardship empowering upper echelon sustainability balance exploitation versus exploration as well as strategic or financial for strategic control the stronger the firm's orientation towards free energy surprise minimization based self-organization the stronger the sustainably sustainability so we talked about this last week a lot I didn't put a lot of extra information in here because we did like the antagonistic neural or that perspective last week um but I did just want to include the summary so Bajan at the end of the chapter summarizes it by saying in summary the board must assume the role of sponsor or enforcer of three categorical imperatives defining the success and sustainability of strategic renewal and firm survival first foster cognitive coupling between and within all hierarchical layers I'm really excited to find out exactly how to do this second Embrace environment as action generated and socially constructed shared meaning and shared expectations which we've talked about a lot today and third the chain The Firm from thinking in terms of antagonisms but Embrace real-time Duality management as a critical determinant of performance and survival um that's pretty much it do you guys have any final thoughts the overtime period the generative AI speculation moment the hashtag 2023 questions that we all have times um so I I we talked about Quantum so I'm leaving us with this one um quote from this chapter because it's just kind of like thrown in there Bijan says classical linear cause effect thinking is no match for Quantum exponentiality I wonder what it means by Quantum exponentiality like are we again talking about superposition are we talking about like because that's like I guess the many states existing at once is that like what he means by exponentiality what do you guys think I think John has a poetic bent in him not sure if we're meant to take this fully literally exponents are used in a lot of quantum mechanics readings just like any other good poem we should jump in with these kinds of questions cool well that's next week um so yeah excited to wrap up this part and just a couple more book streams left for us uh winding down again get in touch if you would like to participate in our final discussion we would really welcome any external perspectives all right thanks see you next time